Syllabus Details

Syllabus ID:	14179
Syllabus Name:	**Software Architecture and Design_Kiến trúc và thiết kế phần mềm**
Course Name English:	**Software Architecture and Design**
Subject Code:	**SWD392**
Learning-Teaching Method:	Social Constructivism,In-class lecture,Inquiry-based Teaching
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 45h contact hours + 145-minute final exam + 102,6h self-study
Pre-Requisite:	SWE201c or SWE202c, PRO192
Description:	This is a course in concepts and methods for the architectural design of software systems of sufficient size and complexity to require the effort of several people for many months. Fundamental design concepts and design notations are introduced. Several design methods are presented and compared, with examples of their use. Students will undertake a term project working in small groups addressing the design of a relatively complex software system.
StudentTasks:	Students must attend at least 80% of contact slots in order to be accepted to the final examination. - Student is responsible to do all exercises given by instructor in class or at home and submit on time - Constantly follow announcements on LMS at https://flm.fpt.edu.vn/ for up-to-date course information regarding assignment submission and feedback on assignments and project work.
Tools:	-Internet -The Rational Software Architect CASE tool -Visual Paradigm, MagicDraw, and Visio
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	377/QĐ-ĐHFPT dated 04/09/2026
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	4/9/2026

4 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	Software Modeling and Design: UML, Use Cases, Patterns, and Software Architectures	H. Gomaa	Cambridge University Press	2011	1st Edition	9780521764148	True	True	False	Books24×7, https://library.books24 bookid=41618
2	UML Distilled: A Brief Guide to the Standard Object Modeling Language	Fowler	Addison-Wesley	2004	3rd edition	978-0321193681	False	True	False	
3	https://gmu.blackboard.com/webct/						False	False	True	

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
4	Design Patterns Elements of Reusable Object-Oriented Software	Erich Gamma, Richard Helm, Ralph Johnson, and John Vlissides	Addison-Wesley Publishing Company	1995			False	True	False	

7 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Know the Fundamentals of Software Design that includes Software Design Process, Concepts, Notations and Methods
2	CLO2	Able to explain the steps in Using COMET/UML, design concepts and multiplicity of associations Collaborative Object Modeling and architectural design method (COMET)
3	CLO3	Able to use artifacts in developing Software Analysis Model : classes and objects,statecharts for state dependent objects, object interaction diagrams for each use case Using AI tools generating UML diagrams
4	CLO4	Able to design overall Software Architecture in developing Software Design Model Analyzing design solutions with AI-driven improvement recommendations.
5	CLO5	Able to design relational database in developing Software Design Model
6	CLO6	Able to explain the structure and document a Design Pattern
7	CLO7	Effectively utilize AI tools (ChatGPT, PlantUML, Copilot) to support software architecture analysis and design, including: suggesting appropriate architectural models based on system requirements, generating UML diagrams (Use Case, Class, Sequence, Statechart...)

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Course Introduction Intro to Software Design (Chapter 1-2)	Offline	CLO1	TU	Slide, Video	SWD392	Read Slide Watch Video Do quizzes and assignments (if any)	
2	Course Introduction Intro to Software Design (Chapter 1-2)	Offline	CLO1	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
3	Course Introduction Intro to Software Design (Chapter 1-2)	Offline	CLO1	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
4	Intro to Software Design (Chapter 3-4) Course Project introduction	Offline	CLO1	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
5	Intro to Software Design (Chapter 3-4)	Offline	CLO1	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
6	Intro to Software Design (Chapter 3-4)	Offline	CLO1	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
7	Requirements and Use Case Modeling (Chapter 5-6) Software Quality Attributes (Chapter 20)	Offline	CLO2, CLO3	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
8	Requirements and Use Case Modeling (Chapter 5-6) Software Quality Attributes (Chapter 20)	Offline	CLO2, CLO3	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
9	Requirements and Use Case Modeling (Chapter 5-6) Software Quality Attributes (Chapter 20)	Offline	CLO2, CLO3	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
10	Requirements and Use Case Modeling (Chapter 5-6) Software Quality Attributes (Chapter 20) . Use ChatGPT to generate use case descriptions based on user requirements. Use GitHub Copilot to generate boilerplate code for logic implementation based on the use case	Offline	CLO2, CLO3	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
11	Analysis Modeling – Static modeling (Chapter 7) Analysis Modeling – Object and Class Structuring (Chapter 8)	Offline	CLO2, CLO3	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
12	Analysis Modeling – Static modeling (Chapter 7) Analysis Modeling – Object and Class Structuring (Chapter 8)	Offline	CLO2, CLO3	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
13	Analysis Modeling – Static modeling (Chapter 7) Analysis Modeling – Object and Class Structuring (Chapter 8)	Offline	CLO2, CLO3	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
14	Analysis Modeling – Static modeling (Chapter 7) Analysis Modeling – Object and Class Structuring (Chapter 8)	Offline	CLO2, CLO3	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
15	Analysis Modeling – Dynamic Interaction Modeling (Chapter 9, 11) Finite State Machines and Statecharts (Chapter 10)	Offline	CLO2, CLO3	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
16	Analysis Modeling – Dynamic Interaction Modeling (Chapter 9, 11) Finite State Machines and Statecharts (Chapter 10)	Offline	CLO2, CLO3	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
17	Analysis Modeling – Dynamic Interaction Modeling (Chapter 9-10-11) Finite State Machines and Statecharts (Chapter 10)	Offline	CLO2, CLO3	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
18	Analysis Modeling – Dynamic Interaction Modeling (Chapter 9-10-11) Finite State Machines and Statecharts (Chapter 10) Generate UML architecture diagrams using PlantUML code.	Offline	CLO2, CLO3	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
19	Introduction to Design Patterns (Material 3)	Offline	CLO6	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
20	Creational & Structural Patterns	Offline	CLO7	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
21	Behavioral Patterns	Offline	CLO8	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
22	Use ChatGPT to suggest appropriate design patterns. Progress test 1 Course Project- On-going Assessment 1	Offline	CLO1, CLO2, CLO3, CLO5, CLO6	TU	Slide, Video		Do presentation	
23	Course Project- On-going Assessment 1	Offline	CLO1, CLO2, CLO3, CLO5, CLO6	TU	Slide, Video		Do presentation	
24	Course Project- On-going Assessment 1	Offline	CLO1, CLO2, CLO3, CLO5, CLO6	TU	Slide, Video		Do presentation	
25	Course Project- On-going Assessment 1	Offline	CLO1, CLO2, CLO3, CLO5, CLO6	TU	Slide, Video		Do presentation	
26	Overview of Software Architecture (Chapter 12) Software Subsystem Architectural Design (Chapter 13)	Offline	CLO4	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
27	Overview of Software Architecture (Chapter 12) Software Subsystem Architectural Design (Chapter 13)	Offline	CLO4	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
28	Overview of Software Architecture (Chapter 12) - Software Subsystem Architectural Design (Chapter 13)	Offline	CLO4	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
29	Overview of Software Architecture (Chapter 12) - Software Subsystem Architectural Design (Chapter 13)	Offline	CLO4	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
30	Designing Object-Oriented Software Architectures (Chapter 14)	Offline	CLO4	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
31	Designing Object-Oriented Software Architectures (Chapter 14)	Offline	CLO4	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
32	Designing Object-Oriented Software Architectures (Chapter 14)	Offline	CLO4	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
33	Designing Object-Oriented Software Architectures (Chapter 14)	Offline	CLO4	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
34	Designing Client/Server Software Architectures (Chapter 15) Client/Server Software Architectures Case Study (Chapter 21)	Offline	CLO4	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
35	Designing Client/Server Software Architectures (Chapter 15) Client/Server Software Architectures Case Study (Chapter 21)	Offline	CLO4	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
36	Designing Client/Server Software Architectures (Chapter 15) Client/Server Software Architectures Case Study (Chapter 21)	Offline	CLO4	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
37	Designing Client/Server Software Architectures (Chapter 15) Client/Server Software Architectures Case Study (Chapter 21)	Offline	CLO4	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
38	Progress test 2 Using ChatGPT and PlantUML suggest applying relevant patterns in the design Course Project- On-going Assessment 2	Offline	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	TU	Slide, Video		Do presentation	
39	Course Project- On-going Assessment 2	Offline	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	TU	Slide, Video		Do presentation	
40	Course Project- On-going Assessment 2	Offline	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	TU	Slide, Video		Do presentation	
41	Course Project- On-going Assessment 2	Offline	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	TU	Slide, Video		Do presentation	
42	Designing Service-Oriented Architectures (Chapter 16) Service-Oriented Architectures Case Study(Chapter 22)	Offline	CLO4	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
43	Designing Service-Oriented Architectures (Chapter 16) Service-Oriented Architectures Case Study(Chapter 22)	Offline	CLO4	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
44	Designing Service-Oriented Architectures (Chapter 16) Service-Oriented Architectures Case Study(Chapter 22)	Offline	CLO4	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
45	Designing Service-Oriented Architectures (Chapter 16) Service-Oriented Architectures Case Study(Chapter 22)	Offline	CLO4	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
46	Designing Component-Based Software Architectures (Chapter 17) Component-Based Software Architectures Case Study (Chapter 23)	Offline	CLO4	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
47	Designing Component-Based Software Architectures (Chapter 17) Component-Based Software Architectures Case Study (Chapter 23)	Offline	CLO4	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
48	Designing Component-Based Software Architectures (Chapter 17) Component-Based Software Architectures Case Study (Chapter 23)	Offline	CLO4	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
49	Designing Component-Based Software Architectures (Chapter 17) Component-Based Software Architectures Case Study (Chapter 23)	Offline	CLO4	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
50	Designing Concurrent and Real-time Software Architectures (Chapter 18) Real-time Software Architectures Case Study (Chapter 24)	Offline	CLO4	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
51	Designing Concurrent and Real-time Software Architectures (Chapter 18) Real-time Software Architectures Case Study (Chapter 24)	Offline	CLO4	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
52	Designing Concurrent and Real-time Software Architectures (Chapter 18) Real-time Software Architectures Case Study (Chapter 24)	Offline	CLO4	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
53	Designing Concurrent and Real-time Software Architectures (Chapter 18) Real-time Software Architectures Case Study (Chapter 24)	Offline	CLO4	TU	Slide, Video		Read Slide Watch Video Do quizzes and assignments (if any)	
54	Progrest test 3 Course Project: Support writing architectural documentation. Generate UML diagrams with example code using PlantUML or Mermaid. Suggest design pattern, architectural Patterns application rules	Offline	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	TU			Do presentation	
55	Course Project - Final Evaluation (Team 1-2-3-4)	Offline	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	TU			Do presentation	
56	Course Project - Final Evaluation (Team 1-2-3-4)	Offline	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	TU			Do presentation	
57	Course Project - Final Evaluation (Team 1-2-3-4)	Offline	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	TU			Do presentation	
58	Course Project - Final Evaluation (Team 5-6)	Offline	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	TU			Do presentation	
59	Course Project - Final Evaluation (Team 5-6)	Offline	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	TU			Do presentation	
60	Course Review	Offline	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	TU			Do presentation	

0 Constructive question(s)

5 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Course Project	on-going	1	25.0%	5	Option 1: 30 slots; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	All LOs	Option 1: N/A; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Option 1: N/A; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Software Design and Modeling	in class, by instructor	1. On-going Assessment: 40% Students work in a project team of 4-5 members on the assignments as appointed by the teacher The works are organized by iterations with the contents and assessments as below (each student is assign to practice full-stack codes on assigned/selected function). Throughout the process, students are encouraged to use AI tools (e.g., ChatGPT, GitHub Copilot), under the following conditions: - Must transparently declare AI usage (e.g., what prompt was used, which part was generated by AI). - Must explain AI-generated output using their own understanding. a. Evaluation 1 (20%) – Requirement Analysis & Modeling Create use case descriptions and UML diagrams (static and dynamic models). AI can assist with modeling, but students must present their reasoning independently. Required Deliverables: AI prompts (if used), UML diagrams (e.g., PlantUML or Visio), written explanations. b. Evaluation 2 (20%) – Software Architecture & Design

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
												Design software architecture, decompose into subsystems, object-oriented design, and database schema. AI can help propose architecture or generate code samples, but students must refine and justify the result. Required Deliverables: Architecture docs, diagrams, code snippets, explanation of AI-assisted parts. 2. Final Project Presentation: 60% Each team presents their integrated software product and individual responsibilities. The individual evaluation includes an additional component for AI usage as follows: - Integrated Software Product Demostration for Software design (30%): Web API , Web Admin, Mobile App... - Mastery of assigned works: weighted 20% - Soft Skills & Working Attitudes: weighted 10% AI Evaluation Rubric (for individual performance – up to 10 points) - Understanding of AI Output - Customization and Enhancement of AI Output - Transparency (Prompt + AI Use Declaration) - Effective and Ethical AI Usage

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
												To be selected to the Final Presentation, each student needs: * All on-going assessments (Evaluation 1 & Evaluation 2) >=5/10 * No cheating during the project progress
2	Progress test	on-going	3	15.0%	>0	Option 1: 30'; ; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	All LOs	Option 1: Multiple Choices; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Option 1: 30; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	All LOs	in class, by instructor	There are 3 progress tests for this course. (5% for each progress test)
3	Final exam	Final exam	2	60.0%	4	145 minutes	All CLOs			All subjects in syllabus	By exam board Manual evaluating	Each part of the final exam must be >=4/10 & Final Result >=5/10.
4	Practical Exam	Final exam	1	20.0%	4	85'	All LOs	Assignment		All studied chapters	by Exam Board	
5	Theory Exam	Final exam	1	40.0%	4	60'	All LOs	Multiple Choices	60	All LOs	by exam board, using computer	The exam questions must be different at least 70% to the previous ones.